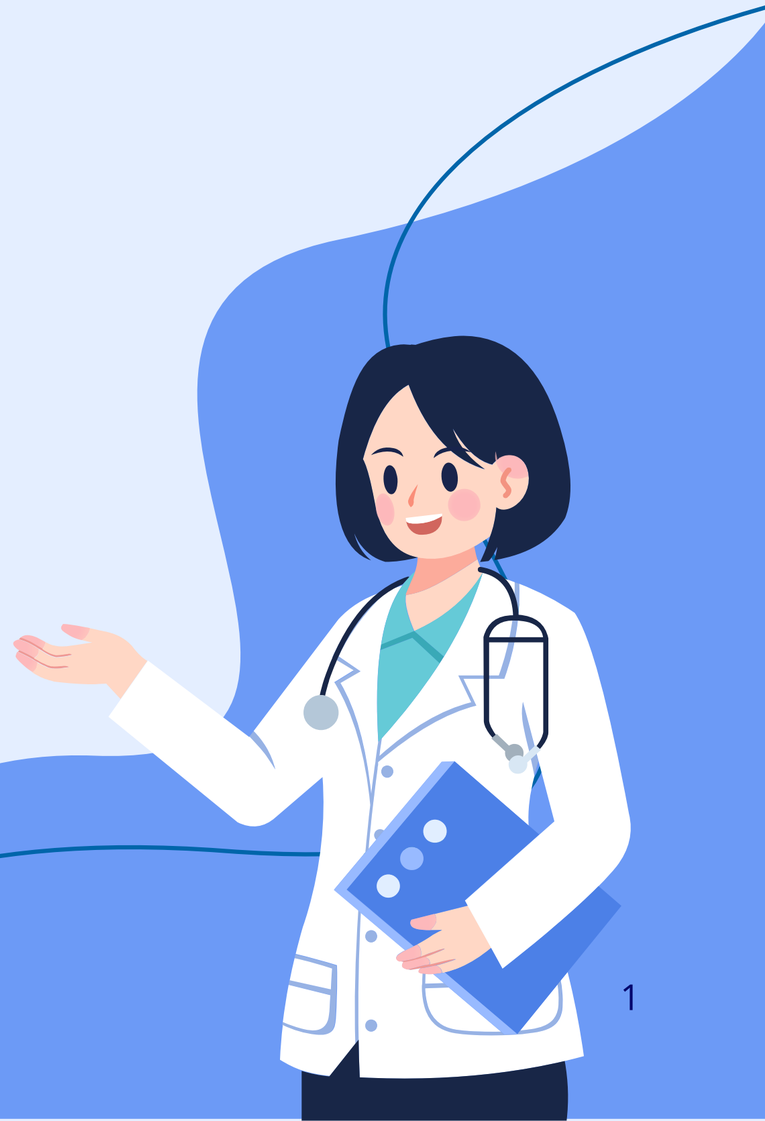
DEEP

Dupilumab Eosinophilia Early Prediction



Checkpoint #1

Team 11: Arianna Biancone, Letizia Brizzi, Luca Campus

Mentors: Sara De Luca, Angelica Urbanelli

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CONTEXT

CHRONIC RHINOSINUSITIS WITH NASAL POLYPS

- Inflammatory disorder of the sinonasal mucosa
- Affects **3%** of individuals in Western countries

DUPILUMAB

- Monoclonal **antibody** approved for treating CRSwNP
- Approved in Italy in 2020

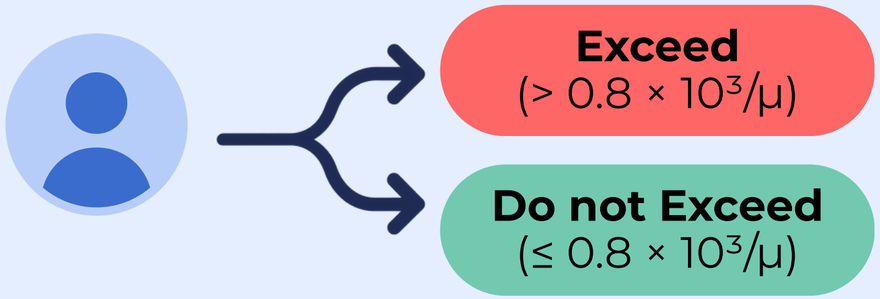
EOSINOPHILIA

- Elevated blood eosinophil counts
- Observed during Dupilumab treatment
- May lead to **treatment discontinuation**

PROJECT GOAL

PRIMARY TASK

Predict **whether** a patient will exceed or not a blood eosinophil threshold of 0.8 × 10^3/μL, based on baseline demographic and clinical features.



EXTENSIONS

Provide an interpretable model highlighting **which** comorbidities and clinical features drive the prediction.

Predict **when** the eosinophil threshold will be exceeded.

VALUE PROPOSITION



Improves **patient safety** and **care quality** through the early identification of potential eosinophilia risk cases, enabling personalized treatments.



Optimizes healthcare **resources allocation** and supports **clinical decisions** by reducing unexpected interventions through predictive analytics.

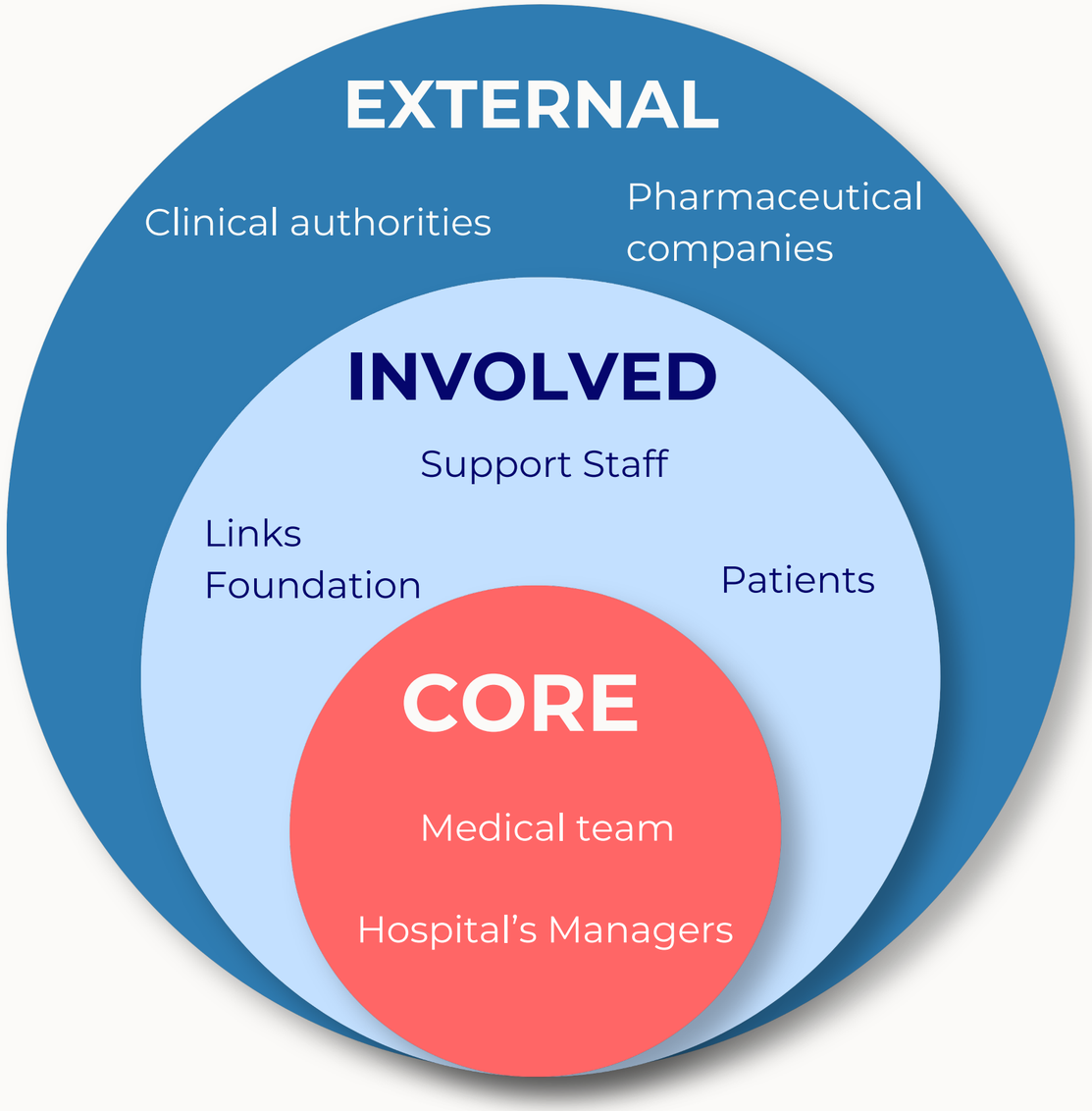
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STAKEHOLDERS' MAP



PERSONAS



DR. STRANGE

ENT Doctor



MS. PEARSON

Resource Administration Manager



DR. STRANGE JOURNEY

	INSERT	CHECK	DECIDE
USER ACTION	Enter demographic and patient **information**	**Review** predicted eosinophilia **risks**	Decide **whether** treatment should be **initiated**
POSSIBLE PAIN	Time-consuming manual data entry	**Complex** data and unclear **predictions**	**Overreliance** on model predictions
OPPORTUNITY	Automate data collection and entry	Provide **intuitive** graphs and risk indicators	Improve model decisions **explainability**



MS. PEARSON JOURNEY

	MONITOR	VERIFY	ORDER
USER ACTION	Check platform's global stathistics and **estimate** demand	**Confirm** estimated demand with clinicians	Place Dupilumab **orders**
POSSIBLE PAIN	**Hard** to **translate** predictions into quantities	Time-consuming coordination	Risk of unexpected stockouts
OPPORTUNITY	**Suggest** order quantities	Allow doctors to flag questionable predictions	Automated safety buffer recommendations

FUNCTIONAL REQUIREMENTS

NEED	FUNCTIONAL REQUIREMENT
R1 Assess the risk of Eosinophilia	**Predict** the risk of **exceeding** the eosinophil **threshold** based on baseline demographic and clinical features
R2 Ensure model explainability	Provide **feature importance** and **explanations** for each individual prediction.

FUNCTIONAL DIAGRAM



DATABASE
- Anagraphic data
- Follow-up visits

PREPROCESSING
- Data cleaning
- Missing values handling

FEATURE EXTRACTION
- Correlation
- Most relevant ones

MODEL
- Logistic Regression
- Clustering

OUTPUT
- Treatment initiation probability (%)

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MODEL RISKS

01

FALSE POSITIVES / FALSE NEGATIVES

Safe treatments may be denied, while risky therapies may be started only to be discontinued.

02

BIAS AND CLASS IMBALANCE

Uneven group representation may lead to biased predictions.

03

OVERFITTING AND LOW ACCURACY

Limited data, many features, and missing values may reduce accuracy and generalizability.



KPI

- **F1-SCORE**

  To assess both **precision** and **recall**

≥ 0.5

- **EXPLORED MODELS**

  Logistic Regression, Random Forest Classifier, Linear SVM etc.

≥ 3

WHAT'S NEXT?

KEY ACHIEVEMENTS

- Problem formalization
- Domain understanding
- **Design** completion

FUTURE STEPS

- Exploratory Data Analysis
- Data **preprocessing**
- Feature selection
- Model **development**

THANK YOU FOR THE ATTENTION!

